

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 8: Prompt Reactivity Feedback Phenomena:

TRANSIENTS AT SMALL TIMES & ASYMPTOTIC TRANSIENTS

Reactor without Delayed Neutrons

Let $N_F(t)$ = number of fissions / cm³-sec at time t

Then, the fission rate at time $(t + l_p)$ will be: $N_F(t + l_p) = k_\infty N_F(t)$

Taking into account that $N_F(t + l_p) \simeq N_F(t) + l_p \frac{dN_F(t)}{dt}$ it follows that: $\frac{dN_F(t)}{dt} \simeq \frac{k_\infty - 1}{l_p} N_F(t)$

The solution is: $N_F(t) = N_F(0) \exp\left(\frac{k_\infty - 1}{l_p} t\right)$

We can also write: $N_F(t) = N_F(0)e^{t/T}$ with $T = \frac{l_p}{k_\infty - 1}$ called *reactor period*

Example: Given that $l_p = 10^{-4}$, if within 1 second the multiplication factor changes from 1.0 to 1.001, then the reactor period will be 0.1s - it means that the reactor will pass through 10 periods in 1 second only.

That is, in 1 seconds the fission rate (power) will increase $\exp[1/0.1] = 22000$ times!!!

Reactor with Delayed Neutrons

Assumptions:

1. *One delayed neutron group due to decay of a single precursor*
2. *An infinite homogeneous thermal reactor*

Time dependent diffusion equation for the thermal neutrons and adding the equation for the precursor concentration:

$$\left| \begin{array}{l} (1 - \beta)k_{\infty}\phi_T + \frac{p\lambda C}{\bar{\Sigma}_a} - \phi_T = l_p \frac{d\phi_T}{dt} \\ \frac{dC}{dt} = \frac{\beta k_{\infty} \bar{\Sigma}_a \phi_T}{p} - \lambda C \end{array} \right.$$

Coupled DEs that must be solved simultaneously

Reactor with Delayed Neutrons - Derivation of Reactivity Eq.

Consider the following situation:

- Critical infinite reactor
- At time t a step-wise change of reactivity is introduced (positive or negative)
- The flux change will be: $\phi = Ae^{\omega t}$
- The change in the precursor concentration will be: $C = C_0 e^{\omega t}$
- Then, the equation for the precursor concentration will become:

$$\frac{dC}{dt} = \frac{\beta k_{\infty} \bar{\Sigma}_a \phi_T}{p} - \lambda C \longrightarrow C_0 = \frac{\beta k_{\infty} \bar{\Sigma}_a A}{p(\omega + \lambda)}$$

- And the time dependent diffusion equation will be:

$$(1 - \beta)k_{\infty} + \frac{\lambda \beta k_{\infty}}{\omega + \lambda} - 1 = \omega l_p \quad \text{or} \quad \underbrace{\frac{k_{\infty} - 1}{k_{\infty}}}_{\text{reactivity}} = \frac{\omega l_p}{1 + \omega l_p} + \frac{\omega}{1 + \omega l_p} \frac{\beta}{\omega + \lambda}$$

$$\rho = \frac{\omega l_p}{1 + \omega l_p} + \frac{\omega}{1 + \omega l_p} \frac{\beta}{\omega + \lambda}$$

Reactivity equation – gives the relation between the reactivity and the reactor period $T=1/\omega$

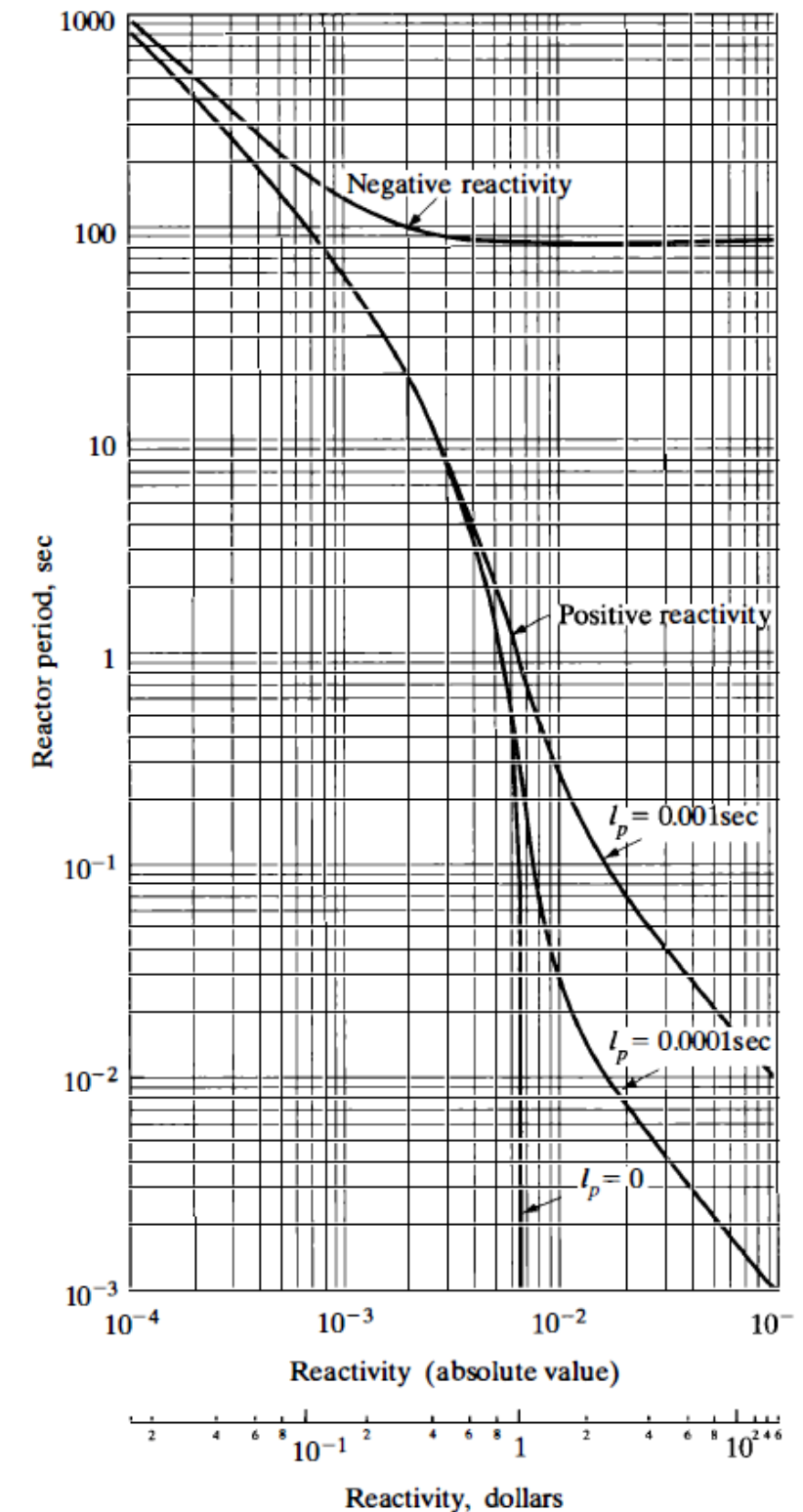
Reactor with Delayed Neutrons

Reactor period for ^{235}U -fuelled reactor as function of positive and negative reactivity:

Example: If the multiplication factor changes from 1.0 to 1.001, the corresponding reactivity will be $\rho = 10^{-3}$

Using that $l_p = 10^{-4}$, the reactor period will be ~ 60 sec

That is, in 1 seconds the fission rate (power) will increase $\exp[1/60] = 1.0168$ times.



Prompt Critical Reactor

Definition: A reactor is called *PROMPT CRITICAL* when it is critical on prompt neutrons alone

→ $(1 - \beta) k = 1$

→ $\rho = \beta$

→ Very short reactor period T

Definition: The amount of reactivity needed to make a reactor prompt critical is known as *a reactivity of one dollar*

Is the dollar of reactivity an absolute unit?

No – because β varies from fuel to fuel.

Nuclide	β (thermal neutron fission)	β (fast fission*)
²³² Th	—	0.0203
²³³ U	0.0026	0.0026
²³⁵ U	0.0065	0.0064
²³⁸ U	—	0.0148
²³⁹ Pu	0.0021	0.0020

*Fission induced by prompt neutron spectrum.

Prompt Jump (Drop) Approximation

Consider a critical reactor that experiences a sudden change in reactivity.

The corresponding change of the neutron flux will be:

one group of delayed neutrons

$$\phi = A_1 e^{\omega_1 t} + A_2 e^{\omega_2 t}$$

six groups of delayed neutrons

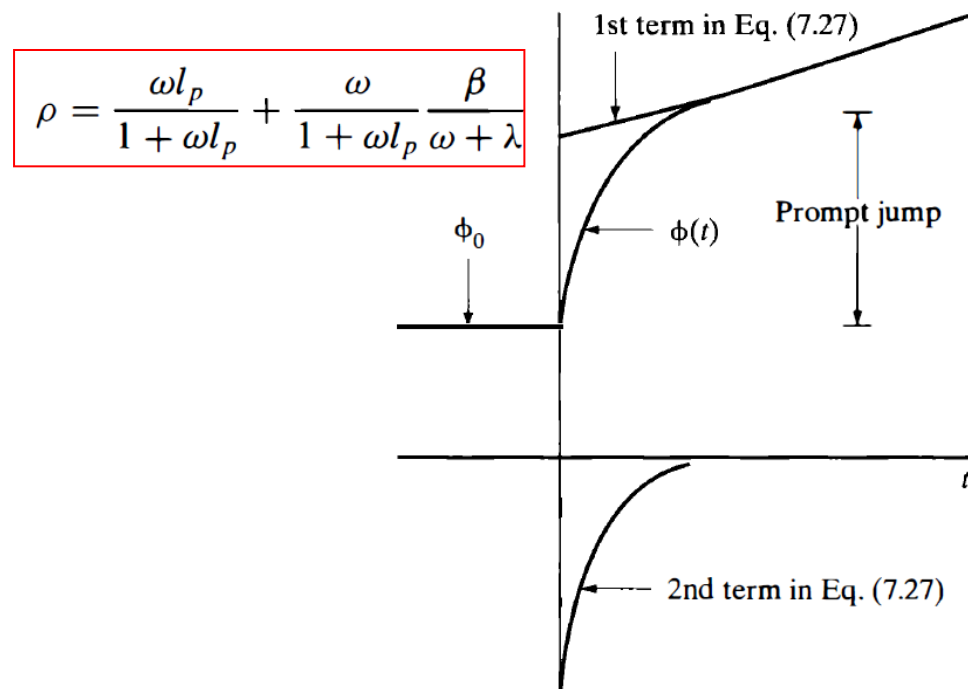
$$\phi = A_1 e^{\omega_1 t} + A_2 e^{\omega_2 t} + \dots + A_7 e^{\omega_7 t}$$

It can be shown that:

- If $\rho > 0 \rightarrow A_2 < 0 \rightarrow \Phi$ is increasing
- If $\rho < 0 \rightarrow A_2 > 0 \rightarrow \Phi$ is decreasing

Note:

- To determine the overall time behavior of the reactor, we should know the level of the initial flux increase/ drop



PJA (Prompt Jump Approximation):

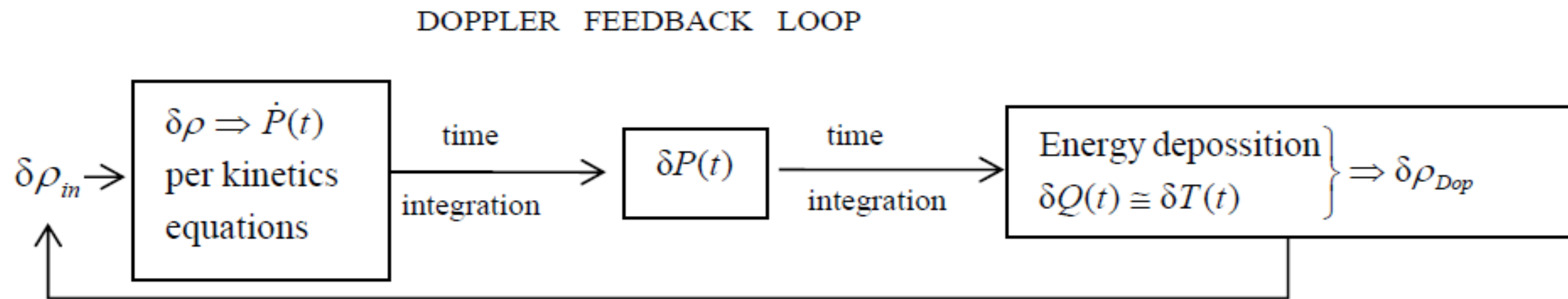
The concentrations of the delayed neutron precursors do not change during the prompt rise/drop in flux

To recall: Prompt Feedback

Doppler Broadening of Resonances

There is only one really prompt feedback effect in thermal as well as in fast reactors, the Doppler reactivity effect.

Doppler broadening of resonances occurs simultaneously with the dissipation of the kinetic energy of the fission products into fuel temperature change.

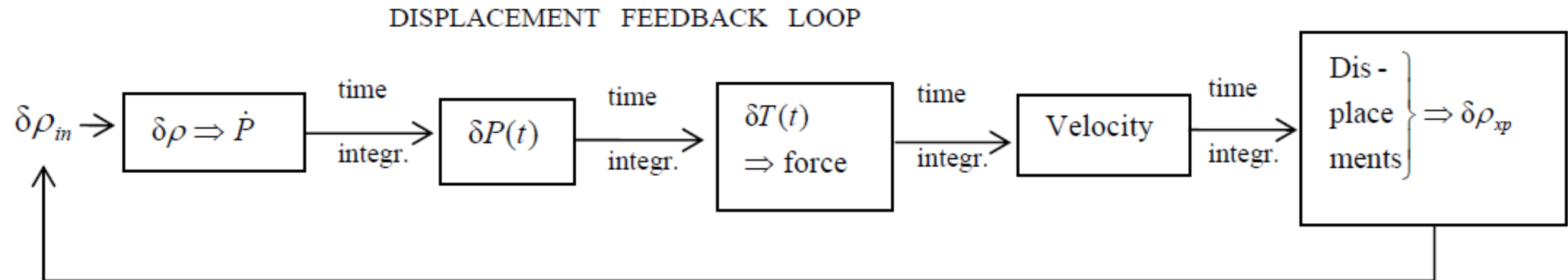


To recall: Prompt Feedback

Axial Expansion of Fuel

The axial expansion of the fuel is the other feedback effect that may be considered as “prompt” in most transients.

The displacement is established almost at the speed of sound in the fuel material (typically ~ 1000 m/s); the corresponding delay is then within ~ 1 ms.



To recall: Prompt Feedback

Rapid transients → adiabatic boundary conditions (no heat transfer).

Models for average fuel temperature rise after the onset of a transients:

Completely adiabatic fuel:

$$\delta T(t) = C_{QT} \int_0^t P(t') dt'$$

Adiabatic fuel for the power increment only
- “linear energy model”:

$$\delta T(t) = C_{QT} \int_0^t [P(t') - P_0] dt'$$

IF we model the heat release out of the fuel with a single time constant, λ_H , **THEN:**

$$\delta Q_{fuel} = \int_0^t [P(t') \exp[-\lambda_H(t - t')]] dt'$$

$$\delta Q_{released} = \int_0^t \delta P(t') \{1 - \exp[-\lambda_H(t - t')]\} dt'$$

$$\delta Q_{total} = \int_0^t \delta P(t') dt'$$

To recall: Prompt Feedback

The temperature increase:

$$\delta T(t) = \frac{1}{c_h} \frac{P_n [W]}{V_f [mm^3]} \int_0^t \delta p(t') dt'$$

The incremental energy in dimensionless units (*“relative full-power seconds”*):

$$I(t) = \int_0^t [p(t') - p_0] dt'$$

$$\delta T(t) = C_{IT} I(t)$$

$$C_{IT} [K / \text{fp} - s] = \left(\frac{\partial T}{\partial t} \right)_{P_n}$$

PROMPT FEEDBACK

In rapid transients, it is convenient to convert the temperature coefficient into an energy coefficient (so far, the reactivity effect of the Doppler broadening of resonances is calculated in the form of a temperature coefficient):

1. For transients described by the flux amplitude $p(t)$

$$C_{QT} \left(\frac{\partial \rho}{\partial T} \right)_{Dop} = \gamma_e^D \left[\frac{\delta \rho}{MW \cdot s} \right] = \text{energy coefficient}$$

$$\text{with } C_{QT} = \frac{\delta T(t)}{\delta Q_{cf}(t)},$$

where Q_{cf} is the energy deposited in the fuel

2. For transients described by the power

$$C_{IT} \left(\frac{\partial \rho}{\partial T} \right)_{Dop} = \gamma^D \left[\frac{\delta \rho}{fp - s} \right] = \text{full - power second coefficient}$$

$$\text{with } C_{IT} = \frac{\delta T(t)'}{I(t)}$$

$$\text{where } I(t) = \int_0^t [p(t') - p_0] dt'$$

PROMPT FEEDBACK

Usually, the total energy coefficient, $\gamma = \gamma^D + \gamma^{exp}$, is being used

IF the fuel is treated **adiabatically** for the power increment

AND

IF the **temperature dependence** of the energy coefficient is neglected

AND

IF first order heat transfer is allowed,

THEN the feedback reactivity is simply given by:

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] \exp[-\lambda_H(t - t')] dt'$$

PROMPT FEEDBACK

The expansion coefficient is essentially independent of temperature → temperature dependence of the total coefficient results primarily from the Doppler effect.

IF the temperature dependence of the temperature coefficient is taken into account,

AND

IF an adiabatic approximation for the increment heat is assumed

THEN feedback reactivity is:

$$\delta\rho(t) = \int_{T_0}^{T(t)} \frac{\partial\rho}{\partial T} dT = \int_0^t \frac{\partial\rho}{\partial T} \frac{\partial T}{\partial t'} dt'$$

where

$$\frac{\partial T}{\partial t'} = C_{IT} [p(t') - p_0]$$

PROMPT FEEDBACK

The temperature dependence of the Doppler coefficient shows approximately a $1/T^x$ behavior:

$$\left(\frac{\partial \rho}{\partial T}\right)_D = \frac{A_D}{T^x}$$

- for thermal reactors $x = 0.5$
 - for fast reactors $x = 1.0$
 - $0.003 \leq A_D \leq 0.009$

The temperature dependence of the Doppler – example:

$A_D = \text{constant} = 0.006$
 $\Delta T_1 = [1000\text{K}; 1900\text{K}]$
 $\Delta T_2 = [1900\text{K}; 2000\text{K}]$

x	$\Delta\rho_1$	$\Delta\rho_2$
0.5	0.14359	0.11191
1.0	0.00385	0.00233
1.2	0.00091	0.00049

Transients in Subprompt-Critical or Subcritical Reactivity Domain

SUBPROMPT-CRITICAL OR SUBCRITICAL REACTIVITY DOMAIN

Initial State: $\rho < \beta$

Transient: Step reactivity insertion of ρ_1

Approximations:

1. Prompt Jump Approximation (PJA)

The PJA is generally a good approximation on reactivity domain $\rho < \beta$ (better, $\rho \leq 0.9 \beta$)

$$\Lambda \approx 4 \times 10^{-7} \text{ s (FBR)}$$

$$\Lambda \approx 10^{-5} \text{ s (LWR)}$$

$$\Lambda \dot{p}(t) \approx 0$$

2. One delayed neutron group: $t \approx 0.2/\bar{\lambda}$

Note that feedback coefficients measured in dollars per full-power second are similar for all oxide fuels.

TRANSIENT AT SMALL TIMES

Given the kinetics equations with feedback reactivity as:

$$\Lambda \dot{p} = \left\{ \rho_1 + \gamma \int_0^t [p(t') - p_0] dt' - \beta \right\} p + \sum_k \lambda_k \zeta_k$$

$$\dot{\zeta}_k = -\lambda_k \zeta_k + \beta_k p$$

Applying PJA leads to:

$$0 = \left\{ \rho_1 + \gamma \int_0^t [p(t') - p_0] dt' - \beta \right\} p + \sum_k \lambda_k \zeta_k$$

with the initial condition of $p(0) = p^0 = \frac{\beta}{\beta - \rho_1} p_0$

Transients analyzed with the PJA may “start” with a pseudo-initial flux p^0 , which eliminates the rapid flux change during the prompt jump.

TRANSIENT AT SMALL TIMES

The flux variations after the prompt jump are slow:

→ we will neglect them by a Taylor series for small t

→ we will investigate only the first two terms of the Taylor expansion of p , ρ , and ζ

$$p(t) = p^0 + p' t + \dots$$

$$\rho(t) = \rho_1 + \rho' t + \dots$$

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] dt'$$

$$\dot{\rho} = \gamma[p(t) - p_0] = \gamma[p^0 - p_0] + \dots$$

$$\zeta_k(t) = \zeta_{k0} + \zeta'_k t + \dots$$

$$\dot{\zeta}_k = -\lambda_k \zeta_k + \beta_k p$$

$$\zeta'_k = -\lambda_k \zeta_{k0} + \beta_k p^0 = \beta_k (p^0 - p_0)$$

*the rate of increase of the precursor density
immediately after the prompt jump*

Then, the rate of increase of delayed neutron source will be: $s'_d = (p^0 - p_0) \sum_k \beta_k \lambda_k = (p^0 - p_0) \beta \bar{\lambda}$ 19

TRANSIENT AT SMALL TIMES

Outcomes:

- *The precursor density and the delayed neutron source increase linearly right after the prompt jump.*
- *In the PJA, the increase in the delayed neutron source causes an instantaneous linear increase in the flux level.*
- *On the other hand, the instantaneous counteraction of the prompt feedback tends to decrease the flux.*

TRANSIENT AT SMALL TIMES

Let p'_{del} and p'_{fb} denote the two parts of the total slope that correspond to delayed neutrons and prompt feedback, respectively:

$$p' = p'_{del} + p'_{fb}$$

Both parts can be obtained by inserting the Taylor expansion into the approximate kinetics equation:

$$0 = \left\{ \rho_1 - \beta + \gamma \int_0^t [p(t') - p_0] dt' \right\} p + \sum_k \lambda_k \zeta_k$$

Using that $s'_d = (p^0 - p_0) \sum_k \beta_k \lambda_k = (p^0 - p_0) \beta \bar{\lambda}$ gives:

$$0 = [\rho_1 - \beta + \gamma(p^0 - p_0)t](p^0 - p't) + \beta p_0 + (p^0 - p_0) \beta \bar{\lambda} t$$

TRANSIENT AT SMALL TIMES

At $t = 0$, the equation reduces to

$$(\rho_1 - \beta)p^0 + \beta p_0 = 0$$

which is satisfied by the definition of p^0 ,

$$p(0) = p^0 = \frac{\beta}{\beta - \rho_1} p_0$$

The slope is obtained from the terms proportional to t , which are given by:

$$0 = [\rho_1 - \beta + \gamma(p^0 - p_0)t](p^0 - p't) + \beta p_0 + (p^0 - p_0)\beta\bar{\lambda} t$$

$$[\rho_1 - \beta]p' + \gamma(p^0 - p_0)p^0 + (p^0 - p_0)\beta\bar{\lambda} = 0$$

TRANSIENT AT SMALL TIMES

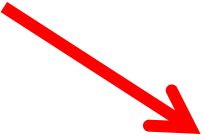
We can show that *the linear increase* of the flux due to the increase in the delayed neutron source is proportional to the product of flux and flux jump:

Omitting the feedback yields the contribution of the increasing precursor population to the slope.

$$[\rho_1 - \beta]p' + \gamma(p^0 - p_0)p^0 + (p^0 - p_0)\beta\bar{\lambda} = 0$$

$$p' = p'_{del} + p'_{fb}$$

$$\gamma = 0$$


$$p'_{del} = \bar{\lambda} \frac{\beta}{\beta - \rho_1} (p^0 - p_0)$$

$$p^0 = \beta p_0 / (\beta - \rho_1) \quad \longrightarrow \quad p'_{del} = \bar{\lambda} \frac{p^0}{p_0} (p^0 - p_0)$$

TRANSIENT AT SMALL TIMES

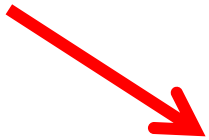
We can show that the counteraction of prompt reactivity feedback leads to *a linear decrease* of the flux:

Omitting the increase of precursors yields the influence of feedback on the variation of the flux after the prompt jump.

$$[\rho_1 - \beta]p' + \gamma(p^0 - p_0)p^0 + (p^0 - p_0)\beta\bar{\lambda} = 0$$

$$p' = p'_{del} + p'_{fb}$$

$$\lambda = 0$$


$$p'_{fb} = \gamma \frac{(p^0 - p_0)}{\beta - \rho_1} p^0 = \frac{\gamma}{\beta} \frac{p^0}{p_0} (p^0 - p_0) p^0$$

TRANSIENT AT SMALL TIMES

Accounting for both feedback and delayed neutrons:

$$p' = \left[\bar{\lambda} + \frac{\gamma}{\beta} p^0 \right] \frac{p^0}{p_0} (p^0 - p_0)$$

1. For small p^0 the slope p' can be positive.
2. For large prompt jumps, the feedback starts to turn the flux down immediately after the prompt jump.

DOPPLER COEFFICIENT IN THE SEFOR EXPERIMENTAL FAST REACTOR

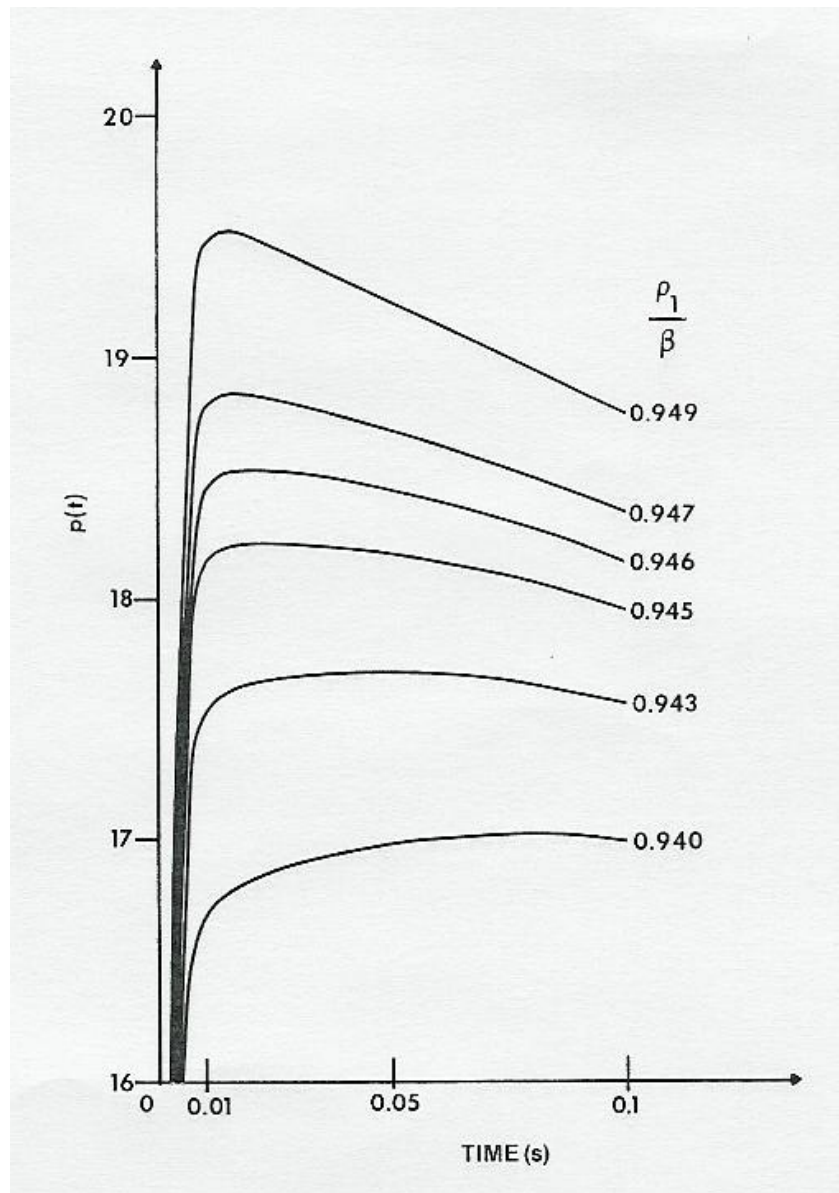
For one special prompt jump, p^{00} , the linear flux change after the prompt jump will vanish and therefore:

$$p' = \left[\bar{\lambda} + \frac{\gamma}{\beta} p^0 \right] \frac{p^0}{p_0} (p^0 - p_0) \quad \longrightarrow \quad -\frac{\gamma}{\beta} p^{00} = \bar{\lambda}$$

If p is the power, then the above equation can be interpreted as the influence of feedback and delayed neutrons on the flux change after the prompt jump is cancelled, i.e., when the energy coefficient times the prompt jump power is equal to the average precursor decay constant.

DOPPLER COEFFICIENT IN THE SEFOR EXPERIMENTAL FAST REACTOR

Computer simulations of transients introduced by a delayed supercritical reactivity step with zero slop after the prompt jump:



The lower curve ($\frac{\rho_1}{\beta} = 0.940$) is still rising after prompt jump; the upper curves are clearly decreasing immediately after prompt jump.

Around $\frac{\rho_1}{\beta} \approx 0.945$, the slope is practically zero after the prompt jump:

The corresponding value of p^{00} is ~18 times the initial flux.

The energy coefficient can be calculated from:

$$\frac{\gamma}{\beta} = -\frac{\bar{\lambda}}{p^{00}}$$

ASYMPTOTIC TRANSIENTS

We can derive an [asymptotic expansion](#) of the approximate kinetic equation $(\{\rho_1 + \gamma \int_0^t [p(t') - p_0] dt' - \beta\} p + \sum_k \lambda_k \zeta_k = 0)$ to find its constant asymptotic solution.

With $\dot{p} = 0$, the kinetic equation in the PJA assumes the form:

$$0 = \frac{\lambda \rho_{as} + \dot{\rho}_{as}}{\beta - \rho_{as}}$$

This requires that the numerator vanish, i.e., $\lambda \rho_{as} + \dot{\rho}_{as} = 0$.

Since $\dot{\rho}_{as} = 0$, the asymptotic time-independent solution of above equation is $\rho_{as} = 0$.

The approach to the asymptotic solution above is describes by the time-dependent solution, i.e.,

$$\rho_{as}(t) \propto \exp(-\lambda t)$$

ASYMPTOTIC TRANSIENTS

Since $\rho_{as} = 0$, the inserted reactivity is asymptotically compensated for by the feedback:

$$-\gamma \int_0^{\infty} [p(t') - p_0] dt' = \rho_1$$

In this equation, the integral must converge, which requires $p(t)$ to converge sufficiently fast toward p_0 .

Thus, a feedback reactivity given by the left side of this equation requires that $p_{as} = p_0$.

Under the assumptions, on which the approximate kinetics equation and its initial condition are based, the prompt feedback forces the power back to its initial value.

ASYMPTOTIC TRANSIENTS

However,

if the feedback is described by $-\gamma \int_0^\infty [p(t') - p_0] dt'$, the asymptotic flux p_{as} cannot be smaller than p_0 :

- ➡ *If $p(t) < p_0$ in a certain time interval, the contribution of this interval to the integral is negative.*
- ➡ *The fuel temperature is decreasing.*
- ➡ *Decreasing fuel temperature together with a negative temperature coefficient leads to a reactivity increase, which is followed by an **increase in the flux**.*

Let's re-evaluate the following three assumptions:

1. PJA

- *Should be accurate asymptotically since it is violated only in the immediate vicinity of fast changes.*

2. One-delay-group kinetics

- *Might be questionable since the kinetics parameters cannot be lumped uniquely into a one-delay-group λ .*
- *λ not even appear in the asymptotic relations: the asymptotic state is a stationary state - the flux level is independent of the precursor decay constant altogether.*

3. No transient heat transfer

- *Violated during the transient and even more so asymptotically.*

ASYMPTOTIC TRANSIENTS

Assumption for no transient heat transfer:

The influence of the heat transfer on the asymptotic flux can be semi-quantitatively investigated by using:

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] \exp[-\lambda_H(t - t')] dt'$$

With $\rho_{as} = 0$, the following equation is obtained:

$$\lim_{t \rightarrow \infty} -\gamma \int_0^t [p(t') - p_0] \exp[-\lambda_H(t - t')] dt' = \rho_1$$

Asymptotically, p_{as} can be inserted as the value of the flux amplitude in performing the limit to large t .

The exponential function practically eliminates the contribution of the integrand, which is far away from the upper limit.

ASYMPTOTIC TRANSIENTS

Assumption for no transient heat transfer (cont.):

Asymptotically, the lower limit “0” is infinitely far away from “ t ”; it can therefore be replaced by $-\infty$

This gives
$$-\gamma \int_{-\infty}^t [p_{as} - p_0] \exp[-\lambda_H(t - t')] dt' = \rho_1$$

or
$$-\frac{\gamma}{\lambda_H} (p_{as} - p_0) = \rho_1$$

Solving for the asymptotic flux amplitude yields:
$$p_{as} = p_0 - \rho_1 \frac{\lambda_H}{\gamma} > p_0$$

Assumption for no transient heat transfer (cont.):

Thus:

- The prompt negative feedback does not shut the reactor down in sense of making it subcritical with an asymptotically vanishing flux.
- The control or shutdown system must be employed to restore the initial power or shut the reactor down.
- Only if p_o is practically zero and λ_H is very small, i.e., if there is practically no heat release, can p_{as} be very small.

ASYMPTOTIC TRANSIENTS

The basic effect of heat transfer is that the flux is not forced back to its initial value → it assumes asymptotically a value above the initial flux.

The numerical calculation of the transient yields the same asymptotic flux value as:

$$p_{as} = p_0 - \rho_1 \frac{\lambda_H}{\gamma} > p_0$$

The formula based on a single delay constant λ_H provides the correct qualitative understanding of why an asymptotically constant flux above the initial flux is obtained.

ASYMPTOTIC TRANSIENTS

Independent of the simplifications employed in describing the heat transfer, the negative reactivity feedback restores criticality.

The inserted reactivity ρ_1 is compensated by a temperature rise:

$$\Delta T = -\rho_1 \left(\frac{\partial \rho}{\partial T} \right)^{-1}$$

- with $\left(\frac{\partial \rho}{\partial T} \right) \approx -0.002 \text{ } \$/\text{K}$, a 0.5\$ reactivity insertion would lead to a temperature rise of 250 K if the control system did not start its counteraction during the transient.
- with $\left(\frac{\partial \rho}{\partial T} \right) \approx -0.002 \text{ } \$/\text{K}$, a 0.95\$ reactivity insertion would lead to a temperature rise of 475 K if the control system did not start its counteraction during the transient.

The temperature rise in the center of the hottest pellet may be substantially larger than that given by the above.

In summary:

1. Prompt negative reactivity feedback reduces a subprompt reactivity insertion to zero: i.e., it restores criticality and does not shut the reactor down.
2. A permanently inserted reactivity is compensated by a permanent temperature rise, given by
$$\Delta T = -\rho_1 \left(\frac{\partial \rho}{\partial T} \right)^{-1}$$
3. The flux approaches asymptotically a constant value. An asymptotic power level higher than the original power is required to sustain a permanent temperature rise.
4. If the core is originally at near zero power and “uncooled”, the Doppler feedback resulting from a “permanent” temperature increase will reduce the neutron flux to near zero and lead to a “shutdown” in this limited sense. However, the asymptotic core would still be critical.

ASYMPTOTIC TRANSIENTS

Example 1: The highest of the curves in the figure ($P_0 = 10 \text{ MW}$)

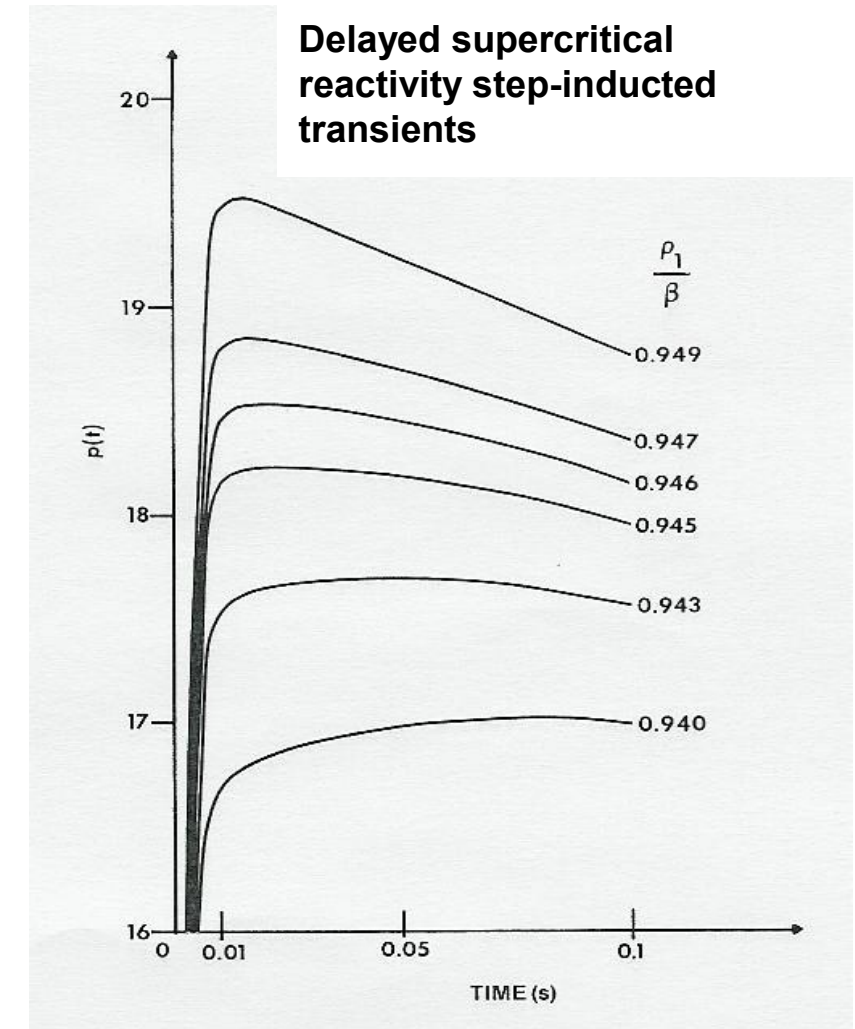
According to

$$-\gamma \int_0^\infty [p(t') - p_0] dt' = \rho_1 \quad \text{and} \quad p_{as} = p_0$$

the transient is supposed to converge to p_0 .

Its integral is approximately given by:

$$\int_0^\infty [p(t') - p_0] dt' = -\frac{\rho_1}{\gamma} = \frac{0.949}{2.15 \times 10^{-3}} \text{ MW} \cdot \text{s} \cong 430 \text{ MW} \cdot \text{s}$$



ASYMPTOTIC TRANSIENTS

Example 2: Subprompt-critical transients with $\rho_1=0.95\%$ (top) and 0.5% (bottom), $p_0=1$

Energy coefficient of: $\gamma = -0.8\$/fp-s$

Reactivity feedback by:

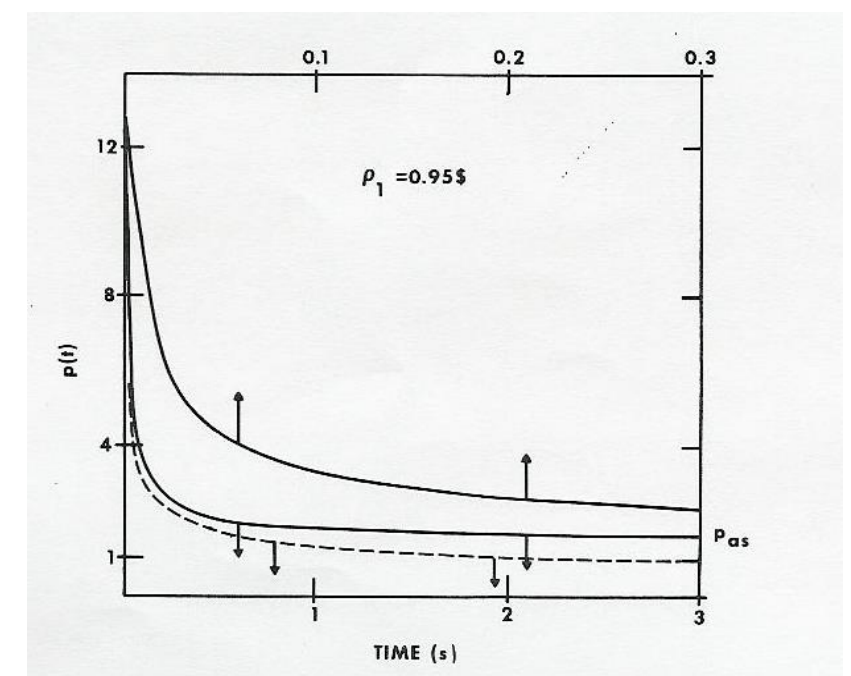
$$0 = \left\{ \rho_1 - \beta + \gamma \int_0^t [p(t') - p_0] dt' \right\} p + \sum_k \lambda_k \zeta_k$$

According to $\int_0^\infty [p(t') - p_0] dt' = -\frac{\rho_1}{\gamma}$ the energy releases is:

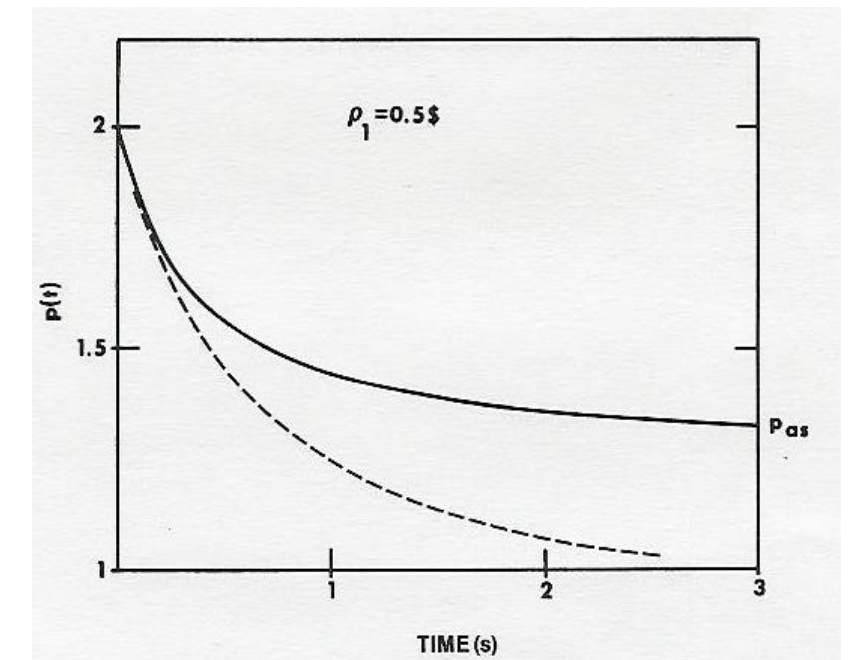
$1.19 fp-s$ if $\rho_1=0.95\%$

and

$0.625 fp-s$ if $\rho_1=0.5\%$



Subprompt-critical transients ($\rho_1=0.95\%$ (top) and 0.5% (bottom), $p_0=1$) with reactivity feedback including first-order heat transient (solid line) and no heat transfer (dashed line)



ASYMPTOTIC TRANSIENTS

Example 2: (cont.)

The dashed lines cross the initial flux levels at ~ 3 s.

The overshooting of the final flux level is due to the following:

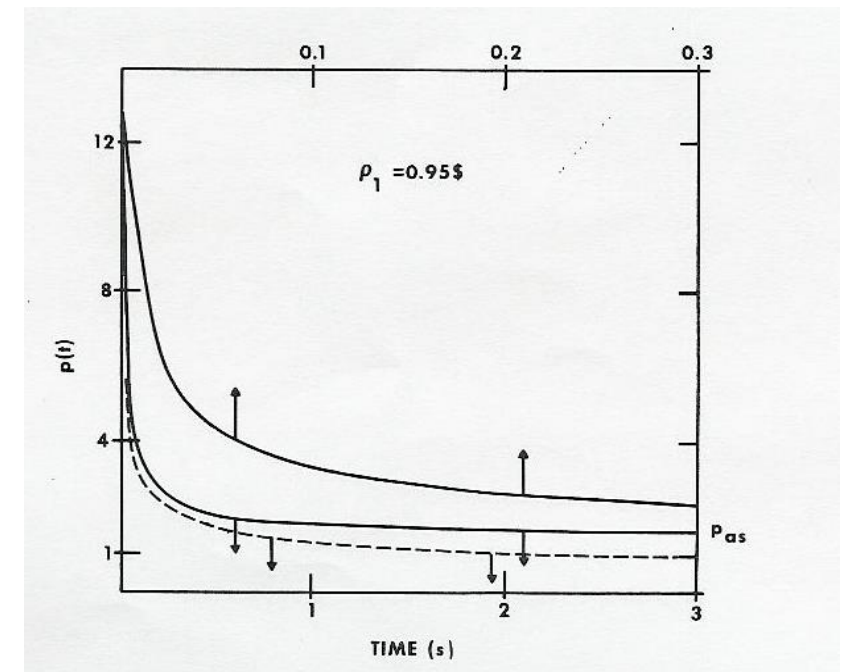
The stationary flux level is given by $p_0 = s_{d0}/\beta$.

This is fulfilled initially and it must hold asymptotically since $p_{as} = p_0$.

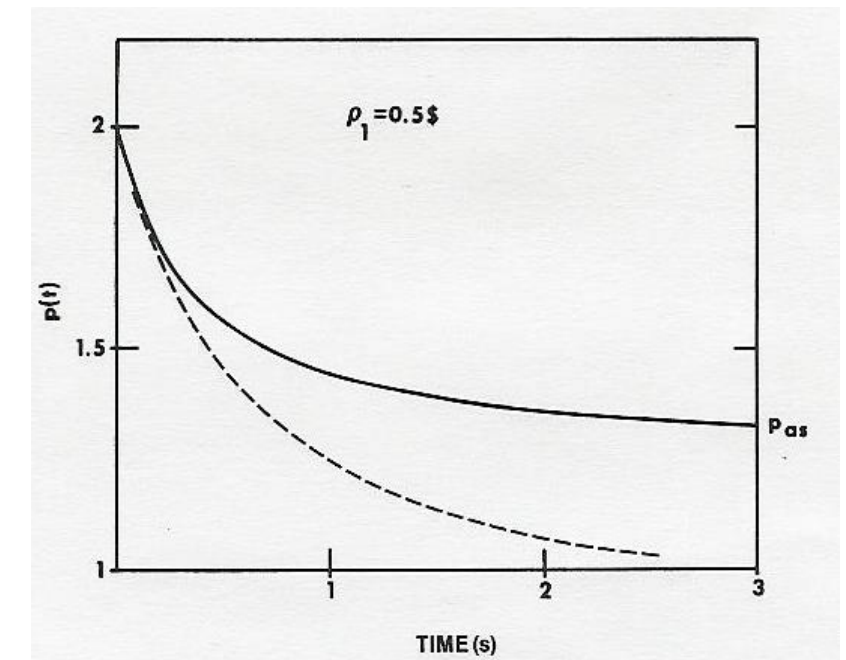
If $p(t) \geq p_0$ and $\rho(t) \geq 0$ during the entire transient, the reactor would be permanently “loaded” with additional precursors:

Then the asymptotic flux should be larger than p_0 in contradiction to $p_{as} = p_0$.

Thus, the flux must be less than p_0 for some period of time to eliminate the additional precursors that are produced in the early part of the transient in order to achieve asymptotically $s_d = s_{d0}$.



Subprompt-critical transients ($\rho_1=0.95$ (top) and 0.5 (bottom), $p_0=1$) with reactivity feedback including first-order heat transient (solid line) and no heat transfer (dashed line)



Next Class

*Superprompt-Critical Excursion Following Step Reactivity
Insertion*